

grey: GENESIS 2.4 orange: the gate blue: added in GENESIS 2.5

SLI script, cell files,
channel prototypes

hsolve SETUP compiles the model
into an opcode program and
coefficient arrays

CPU hsolve
(GENESIS 2.4 solver,
with the fixes)

no

Accelerator gate: chanmode 4 or 5?
Every opcode supported on the device?
Requested precision available?

yes

CUDA or OpenCL,
fp32 (default) or fp64,
chosen at run time
(GENESIS_GPU_PRECISION)

per step

channel update and solve,
one launch per step
(spiking networks)

activations

host, every step: synaptic
events and plasticity
(double); spike flags back

multiloop

isopotential cells,
 K steps per launch
chip_channel_multiloop

multiloop and tree loop: state uploaded once, membrane
potentials returned once per batch of K steps

tree loop

dendritic trees, one thread
per tree, K steps per launch,
CUDA Graphs by default
hines_tree_eliminate